

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why software tests fail — and hardware attestation wins

Problem statement

Commercial LLM APIs are black boxes: users cannot verify the served model matches the advertised model.

This talk shows why software-only audits are unreliable and motivates Trusted Execution Environments for verifiable inference.

The Model Substitution Problem: Economic Incentives Drive Covert S

Formal framing

Hypothesis test over output distributions

$$H_0: P_{\text{actual}}(y|x) = P_{\text{spec}}(y|x)$$

$$H_1: P_{\text{actual}}(y|x) \neq P_{\text{spec}}(y|x)$$

Goal: distinguish specified vs actual model behavior from API samples

Why the incentive exists

Black-box APIs hide implementation details, while GPU costs motivate cheaper substitutes that preserve user-perceived quality.

Substitution types (examples)

 **Smaller model**
e.g., 8B instead of 70B

 **Quantized variant**
FP8/INT8 instead of FP16

 **Fine-tuned / updated**
Different weights, same "brand"

 **Different model family**
Entirely different architecture

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard

Model family	Prec.	MMLU	GSM8K	MATH	GPQA D.
Llama-3-8B-Instruct	Orig	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct	FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	Orig	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct	FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	Orig	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it	FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	Orig	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct	FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	Orig	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3	FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

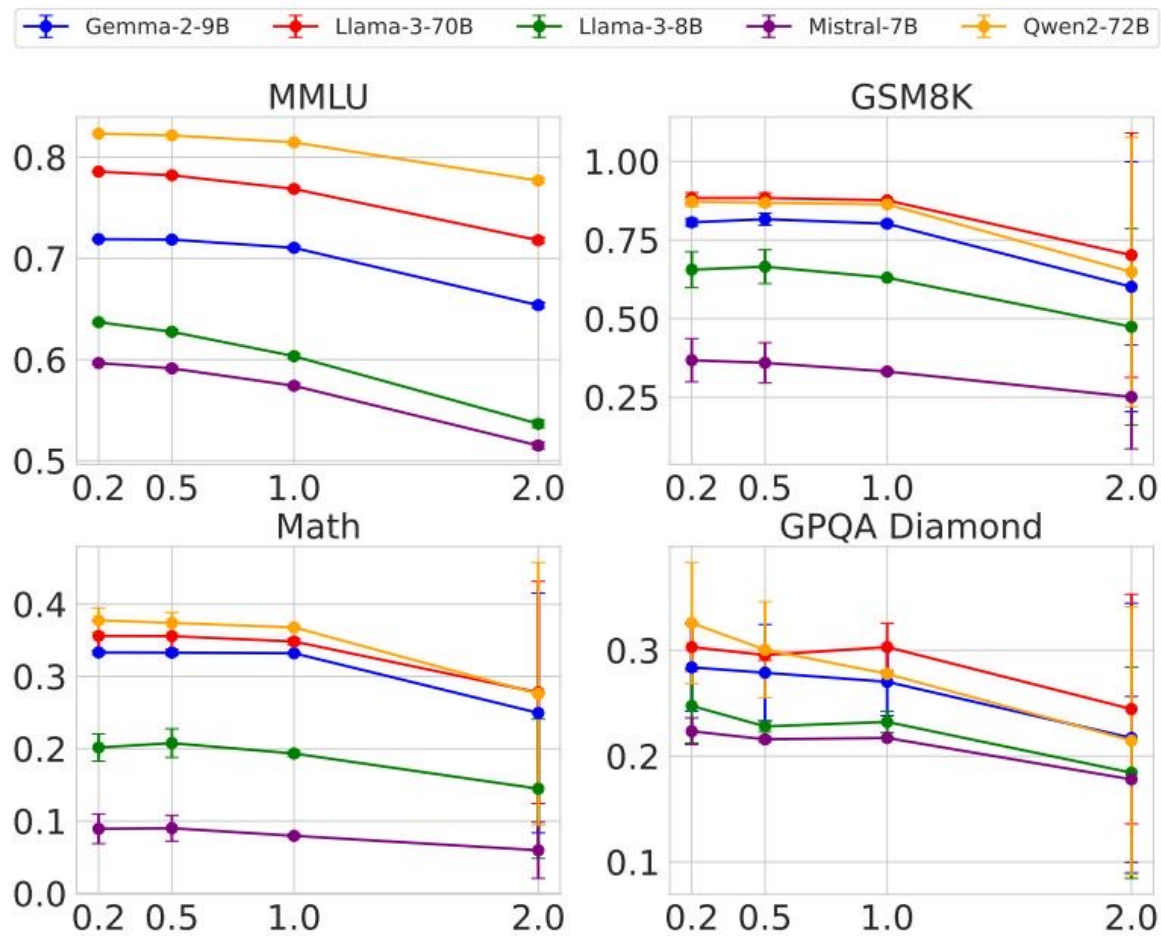
Observation

For most models, FP8 scores fall within error bars (esp. MMLU, GSM8K) → subtle substitution signal.

Implication

Benchmark-only audits are fragile.

Higher Temperature Amplifies Performance Gaps but Increases Noise



Interpretation

- Temperatures: 0.2, 0.5, 1.0, 2.0
- Higher T degrades accuracy for all models
- Substitution signal becomes noisier

Practical takeaway

Increasing temperature can obscure detection: gaps may narrow or become dominated by sampling noise.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Figure: log-probability variation across frameworks/hardware (Llama-3-8B)

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Assumption

Logprob fingerprints are stable for a fixed model and prompt.

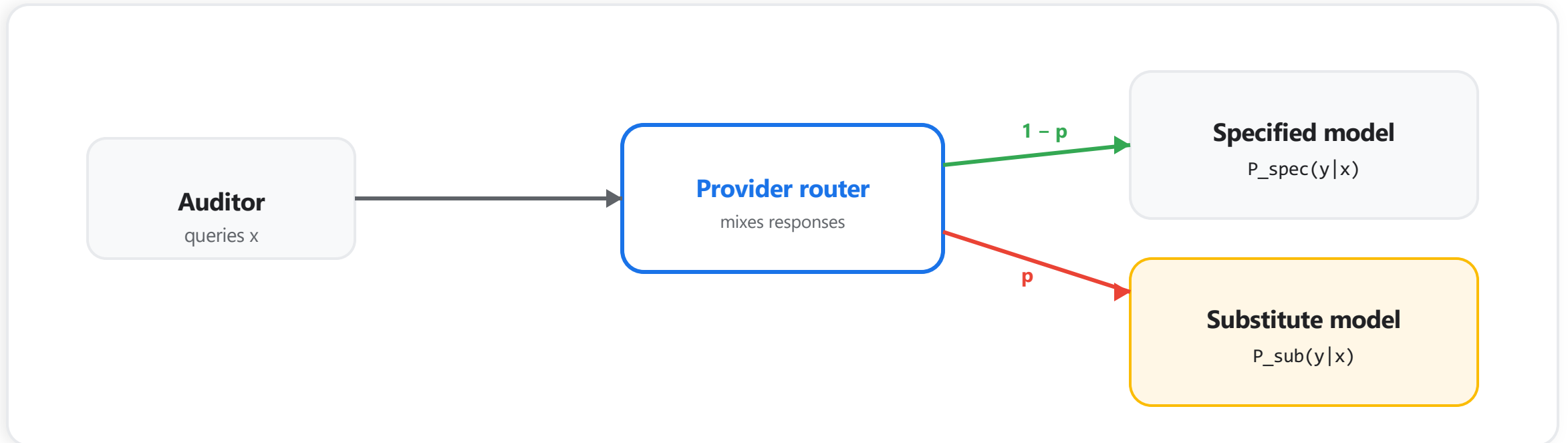
Reality

Same model shows different logprobs across software versions and GPUs.

Consequence

Differences become non-diagnostic: infra drift vs substitution.

Randomized Routing at Low Rates Makes Statistical Detection Query-Pro



Why detection blows up

As $p \rightarrow 0$, the mixed distribution approaches the specified model's distribution.
Software-only tests need impractically large query budgets to detect rare substitution.

$$P_{\text{mix}} = (1-p) \cdot P_{\text{spec}} + p \cdot P_{\text{sub}}$$
$$p \downarrow \Rightarrow P_{\text{mix}} \rightarrow P_{\text{spec}}$$

Software-Only Auditing Methods Are Fundamentally Unreliable

Method type	Weakness	Practical limitation
Text-based statistical tests (e.g., MMD on outputs)	Requires impractically large query budgets; subtle substitutions are hard to separate.	Query-prohibitive under realistic budgets; small deltas often within noise/error bars.
Log-probability fingerprinting	Defeated by production inference nondeterminism across software versions and GPU hardware.	Logprob differences become non-diagnostic: infra drift is indistinguishable from substitution.
Adversarial countermeasures (randomized routing, benchmark evasion)	Adaptive behavior undermines any fixed audit; the provider can shape what the auditor sees.	Providers can route “audit-like” queries to the genuine model while substituting normal traffic.

Structural failure modes: sample complexity + nondeterminism + adaptive adversary

Trusted Execution Environments Provide Provable Cryptographic Model

TEE-attested inference (concept)

1

Load code + weights in enclave

Hardware protects memory; prevents tampering

2

Produce cryptographic attestation

Proof binds runtime to (model weights, code, config)

3

Verifier checks attestation

Audit becomes “verify proof” , not “guess from outputs”

Deployable today

e.g., NVIDIA confidential computing on H100 GPUs

Why TEEs win

Provable

Cryptographic attestation of code + weights

Robust

Not vulnerable to output noise or adaptive routing

Actionable

Modest overhead; deployable in cloud today

Key Takeaways: Close the Verification Gap with Hardware Attestation

- Model substitution is economically motivated and hard to audit via software-only signals.
- FP8 quantization barely changes benchmark scores → subtle substitution fingerprints.
- Inference nondeterminism breaks log-probability fingerprinting in production.
- Randomized substitution makes statistical detection query-prohibitive.
- **TEEs enable verifiable inference with cryptographic attestation — viable today.**

Call to action: Treat integrity as infrastructure—demand attestation for high-stakes LLM APIs.